

Biography

Dr. Ying Lu is a Professor at the Institute of Physics, Chinese Academy of Sciences. Her research lies at the interface of physics and life sciences, where she employs ultrahigh-precision single-molecule fluorescence techniques to unravel the molecular mechanisms underlying fundamental life processes. Her current research includes two main areas: (1) Complex dynamics of biological membrane systems—characterizing the stochastic dynamics, transport processes, and phase behavior of biological membranes and associated membrane proteins from *in vitro* to *in vivo* environments at the single-molecule level; and (2) Deformation and transport in artificial cell membranes—investigating the statistical physics of membrane protein-mediated membrane deformation and the regulation of transmembrane transport.